

Mini Project Report

SRM INSTITUTE OF SCIENCE AND TECHNOLOGY

College of Engineering and Technology

Department of Electronics and Communication Engineering

21ECC101J / Electronic System and PCB design

II Semester, 2023- 24(Even Semester)

Lab code & Title : 21ECC101J & Electronic system and PCB design

Year & Semester : 1 & II

Project Title : Battery voltage indicator

Lab Supervisor :

Team Members Registration No & Name:

Reg. No & Name	Deshna	Shreya	Madhumitha	Nithin
Mark split up ↓	RA2311053010050	RA2311053010056	RA2311053010057	RA2311053010060
Novelty of the project (5 marks)				
Circuit Schematic (10 marks)				
PCB CAM file (SS/DS) (10Marks)				
PCB Fabrication and Assembly (Hardware) (15 marks)				
Project Report (10 Marks)				
Total (50 Marks)				

REPORT VERIFICATION

Date:

Signature of Lab Supervisor

Mini Project Title	Battery voltage indicator
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OBJECTIVE

The objective of this project is to develop a battery voltage indicator that serves as a reliable tool for monitoring the voltage level of batteries used in electronic devices. By accurately displaying battery voltage, the indicator aims to facilitate proactive battery management, thereby extending battery life and optimizing device performance. The project seeks to combine simplicity, affordability, and effectiveness to create a versatile indicator suitable for a wide range of applications..

ABSTRACT

This project presents the design and implementation of a battery voltage indicator intended to monitor the voltage status of batteries in electronic devices. Through the utilization of standard electronic components and a straightforward circuit design, the indicator offers a practical solution for users seeking to monitor battery health. Testing and evaluation of the indicator demonstrate its ability to provide accurate voltage readings and timely feedback, enhancing the usability and longevity of battery-powered devices.

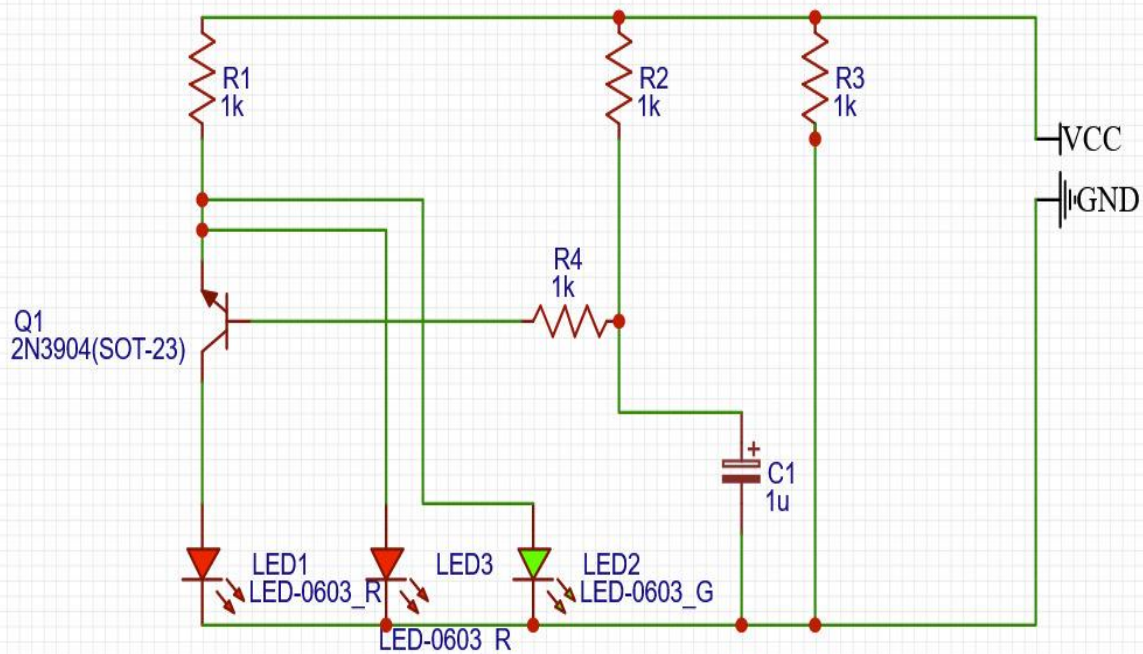
INTRODUCTION

Battery voltage indicators play a crucial role in ensuring the optimal operation of electronic devices by providing users with real-time information about battery status. This project introduces a battery voltage indicator designed to meet the growing demand for reliable battery monitoring solutions. By offering a simple yet effective means of monitoring battery voltage, the indicator aims to empower users to make informed decisions regarding battery usage and maintenance.

SOFTWARE REQUIREMENT/DESCRIPTION

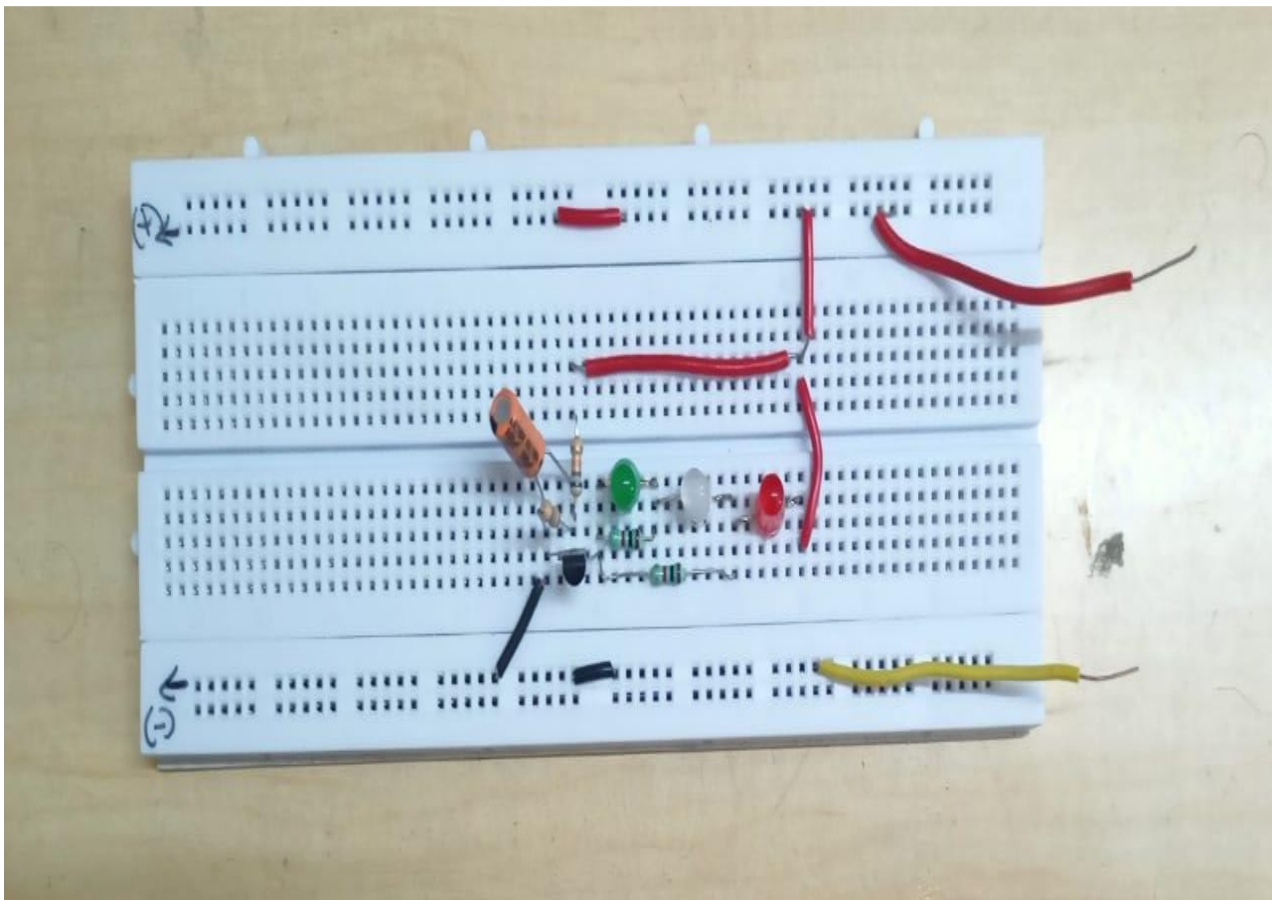
Software requirements encompass the vital specifications and functionalities necessary for a software application to fulfill its intended purpose effectively. EasyEDA, a versatile cloud-based electronic design automation tool, presents a comprehensive platform tailored for designing and simulating electronic circuits and printed circuit boards (PCBs). With its intuitive interface and collaborative features, EasyEDA streamlines the entire design process, from schematic capture to PCB layout, fostering seamless teamwork among engineers and hobbyists alike. Its rich library of components, coupled with powerful simulation capabilities, empowers users to prototype and validate their designs with precision and efficiency. EasyEDA's web-based architecture ensures accessibility across various devices, facilitating flexibility in design workflows. Overall, EasyEDA stands as a robust solution, merging simplicity with sophistication to cater to the diverse needs of electronic design enthusiasts and professionals.

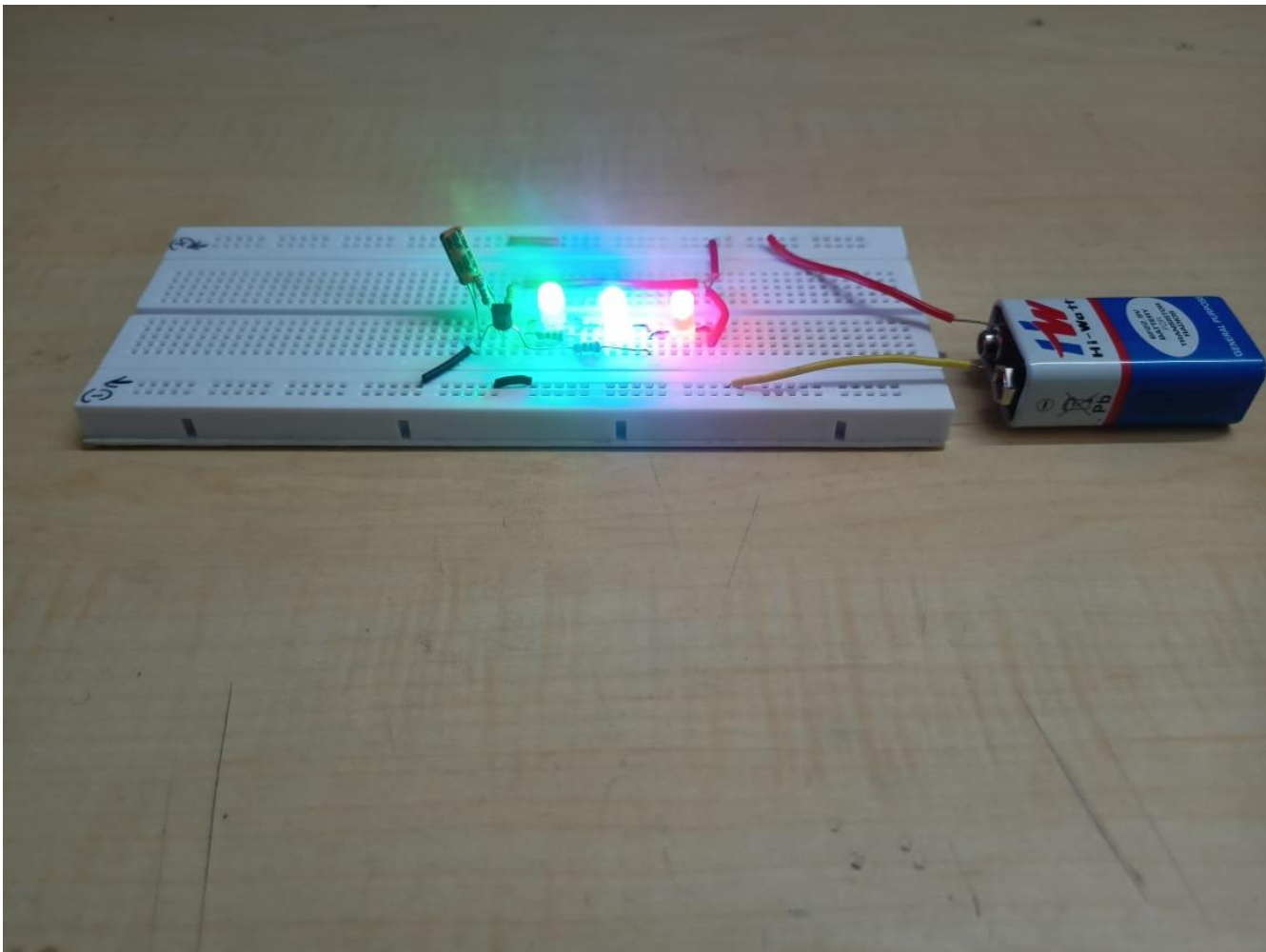
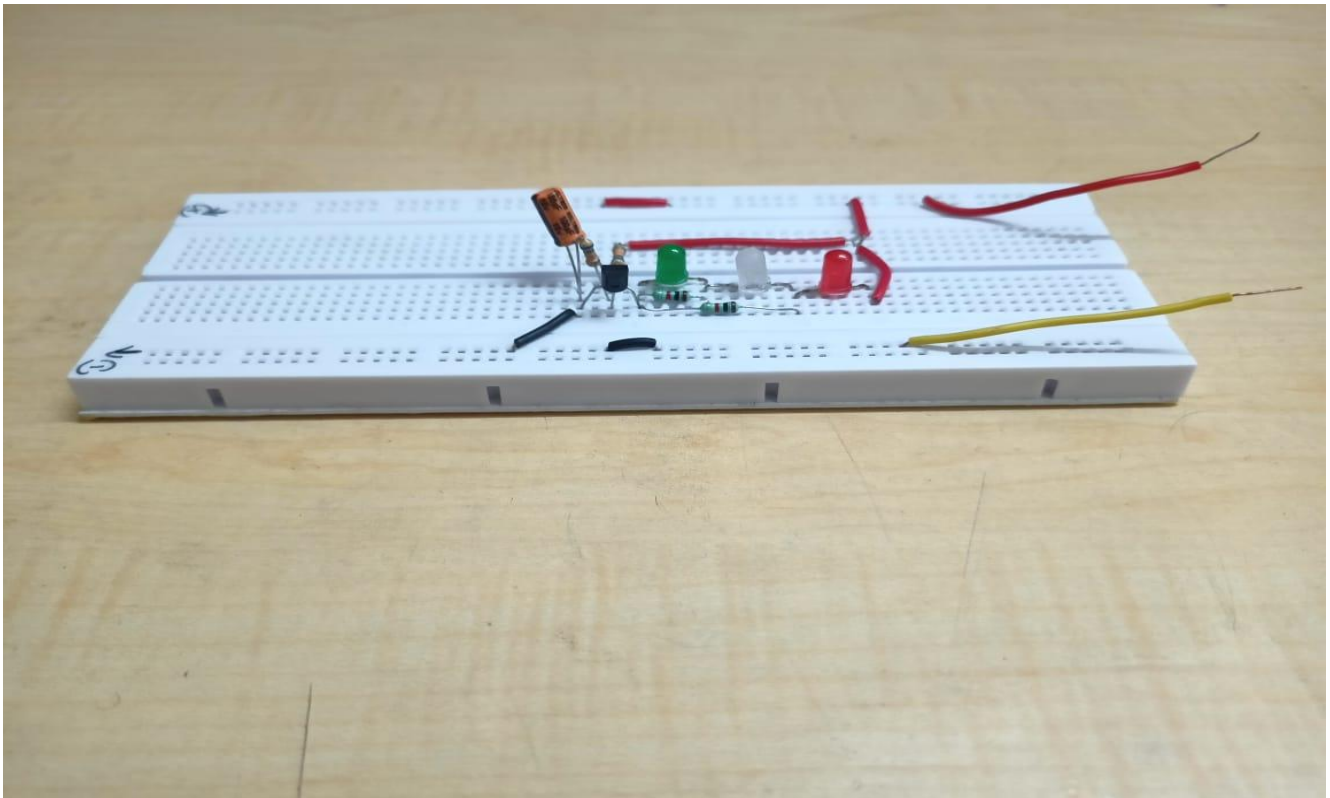
CIRCUIT DIAGRAM AND DESCRIPTION



A battery voltage indicator circuit diagram typically consists of several key components arranged in a specific configuration. At its core, there is usually a voltage divider network comprising resistors connected in series across the battery terminals. This divider network taps off a fraction of the battery voltage proportional to the battery's charge level. A voltage comparator, often implemented with an operational amplifier (op-amp), compares this scaled voltage with a reference voltage level set by a voltage reference component such as a zener diode or a precision voltage regulator. Depending on the comparison result, the comparator outputs a logic signal to drive indicator elements like LEDs or an LCD display, indicating the battery's voltage status. Additionally, the circuit may include filtering components such as capacitors to stabilize the voltage readings and reduce noise. Overall, the battery voltage indicator circuit provides a visual or digital representation of the battery's charge level, enabling users to monitor its status accurately.

FABRICATED PCB ASSEMBLY IMAGE AND ITS OUTPUT MEASUREMENT





RESULTS AND DISCUSSION

Testing of the battery voltage indicator confirms its ability to accurately measure and display battery voltage within the specified range. The indicator exhibits rapid response times and clear, intuitive voltage readings, making it easy for users to interpret battery status. Overall, the results demonstrate the functionality and reliability of the battery voltage indicator in practical applications, validating its effectiveness as a battery monitoring solution.

The performance of the battery voltage indicator meets the project objectives, offering users a simple yet efficient tool for monitoring battery health. While the indicator demonstrates satisfactory accuracy and responsiveness, opportunities for further refinement exist. Future iterations of the project could explore enhancements such as low battery alerts or adjustable voltage thresholds to cater to specific user needs. Additionally, integrating wireless connectivity options or exploring alternative display technologies could enhance the versatility and functionality of the indicator..

CONCLUSIONS

In conclusion, the battery voltage indicator project successfully achieves its goal of providing a practical solution for monitoring battery voltage in electronic devices. The indicator's simplicity, affordability, and effectiveness make it a valuable tool for users seeking to optimize battery usage and device performance. By enabling proactive battery management, the indicator contributes to extended battery life and enhanced user experience. Future developments may focus on incorporating additional features and capabilities to further address user needs and technological advancements.

REFERENCES

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